UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	:	CSC429
EXAMINATION	:	NOVEMBER 2026
TIME	:	

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC429 mock paper (November 2026). Generated answers show working. Real past-paper questions are reproduced with their source reference; no answer is invented where the original scheme is unavailable.

2. Generated marking points state the expected answer and the allocation of marks. Real-bank questions retain their source paper and page reference.

3. Part A = 4 x 5 + 6 x 10; Part B = 1 x 20; total = 100 marks.

4. Generated slots: 2. Real-bank slots: 9. Placeholders: 0. Questions and answers are for local study use only.

5. Structure derived from 20 unique CSC429 past papers; every real question remains traceable to its source page.

FOR MARKERS' USE ONLY

This marking scheme consists of 5 printed pages

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PART A — MARKING GUIDE

QUESTION A1 (5 marks)

Topic: Parallel and multiprocessor architecture

a) How classification of parallel processor is constructed?
b) Give ONE (1) example for each parallel processor architecture.

Source: CSC429 (8).pdf, page(s) 6

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A2 (5 marks)

Topic: Computer generations and history

The Difference Engine was a mechanical computer designed by Charles Babbage in the 19th century. It was one of the earliest examples of a mechanical calculator, built to automatically compute mathematical tables for naval navigation.
a) What device was the successor of the Difference Engine?
b) Describe the difference in function between the Difference Engine and Mechanical Calculating Machine.
c) Briefly explain the available technology that existed in the computer generation of Difference Engine.

Source: CSC429 (17).pdf, page(s) 2

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A3 (5 marks)

Topic: Data representation and number systems

Convert 144010 to binary. Show your complete workflow.

Answer: 144010 = 101101000002

- *1 mark for identifying the target base.*
- *1 mark for a correct repeated-division or place-value method.*
- *1 mark for the correct digit sequence.*
- *1 mark for the correct base subscript.*
- *1 mark for a complete, consistent workflow.*

QUESTION A4 (5 marks)

Topic: Instruction formats and addressing modes

Given the following Figure 2.
Instruction
Opcode
Address A
Memory
Operand
Figure 2
Label the addressing technique illustrated in Figure 2.
a)
Explain how the addressing technique in 9(a) is used.
b)
List ONE (1) advantage and ONE (1) disadvantage of the addressing technique in

Source: CSC429 (1).pdf, page(s) 5

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A5 (10 marks)

Topic: Computer architecture and CPU organisation

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a) Differentiate between organization and architecture.
b) Explain the function of THREE (3) Central Processing Unit (CPU) components.

Source: CSC429 (6).pdf, page(s) 3

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A6 (10 marks)

Topic: Computer architecture and CPU organisation

a) Compare system interconnection and CPU interconnection in terms of:
i) function
ii) physical appearance.
b) Describe the function of the THREE (3) different bus types in a computer system.

Source: CSC429 (16).pdf, page(s) 3

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A7 (10 marks)

Topic: Memory hierarchy and cache

a) Identify the correct access method based on the following scenario
Access method
Scenario
files.
Readi
l
c file from a folder.
Accessi
a
files line
line.
Streami
e file.
an index in a la
AS ecific file block usi
Accessin
toas
in the video.
cific frame or a timestam
Jum
tn
b) Rank the following memory types based on the most to the least storage capacity'
Main memory, Tape, Ll Cache, Hard drive, ReBister, L3 Cache
c) Discuss the term volatile from the perspective of computer memory.

Source: CSC429 (20).pdf, page(s) 3

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A8 (10 marks)

Topic: Input/output and interrupts

Keyboard is the most common means for computer interaction.
a) Explain TWO (2) types of characters that keyboard provides.
b) Describe THREE (3) categories of input and output module.
Give an appropriate
example for each category.

Source: CSC429 (3).pdf, page(s) 4

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

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QUESTION A9 (10 marks)

Topic: Input/output and interrupts

a) Differentiate between interrupt-driven 1/0 and programmed 1/0.
b) Discuss ONE (1) benefit of programmed 1/0.
c) Discuss TWO (2) disadvantages of multiple interrupts.

Source: CSC429 (7).pdf, page(s) 5

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A10 (10 marks)

Topic: Data representation and number systems

Answer both parts for the same expression. Assume an 8-bit byte.
Expression: 30 - (-25)

a) Calculate 30 - (-25) using sign and magnitude.

Answer: left 30 -> 0 0011110; right -25 -> 1 0011001; 30 - (-25) = 55 -> sign-magnitude 0 0110111

- *1 mark for operand representations.*
- *1 mark for sign handling.*
- *1 mark for magnitude arithmetic.*
- *1 mark for the result sign bit.*
- *1 mark for the final byte.*

b) Calculate 30 - (-25) using 1's and 2's complement.

Answer: left 30 -> 1's 00011110, 2's 00011110; right -25 -> 1's 11100110, 2's 11100111; 30 - (-25) = 55 -> 1's 00110111, 2's 00110111

- *1 mark for the positive operand in binary.*
- *1 mark for the 1's complement of the negative operand.*
- *1 mark for adding 1 to form the 2's complement.*
- *1 mark for the correct 8-bit result.*
- *1 mark for converting the result back to decimal.*

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PART B — MARKING GUIDE

QUESTION B1 (20 marks)

Topic: Assembly-language programming

Question 1 is based on Figure 2 and Table 1.

y = A -
2C
* D
B + A2

Figure 2

Write an appropriate program using correct assembly language notations to compute the equation shown in Figure 2 using instructions for one-address, two-address, and three-address. The available instructions are provided in Table 1. The register labeled A, B, C, and D are fixed, while registers x, y, and z can be used temporarily. The final answer must be stored in register y.

Table 1 : Machine instruction

Three-address	One-address	Two-address
LOADp		
MOV (p,q)		
ADD (p.q.r)		
STORp		
ADD (p,q)		
SUB (p,q,r)		
ADDp		
SUB (p,q)		
MUL (p.q.r)		
SUBp		
MUL (p,q)		
DIV(p,q,r)		
MULp		
DIV(p,q)		
DIVp		

Source: CSC429 (18).pdf, page(s) 6

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

END OF MARKING SCHEME

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